

No.	Answer	Solution
19	D	$\sin \theta = \frac{\lambda}{b}, \quad \tan \theta = \frac{\frac{1}{2}x}{D}$ For small angle, $\sin \theta \approx \tan \theta$ $\Rightarrow \frac{\lambda}{b} = \frac{\frac{1}{2}x}{D}$ $\text{Slit width } b = \frac{2\lambda D}{x} = \frac{2cD}{fx}$ The diagram illustrates single-slit diffraction. On the left, a vertical slit of width b is shown. Light rays from the top and bottom edges of the slit travel at angles θ to the central axis. These rays hit a vertical screen at a distance D to the right. The central maximum is centered on the axis, with its total height labeled x. The first minima are located at a distance $x/2$ from the center. The intensity profile is shown as a curve with a central peak and two side lobes.
20	C	Q